FAQs – SFM-03

1. Are stock prices mean reverting?

   Stock prices usually follow a random walk. However, there are periods when they don't. During such times prices can trend upwards or downwards, show choppy behavior, or keep hovering in a narrow range. When prices stay within a range, we say that they are showing mean-reverting behavior.

2. Can you provide some guidance on how we can build a realistic model with the help of a random walk model and Monte Carlo simulations?

   We can combine both models and produce the price simulations. The price range produced by the simulations can then be back-tested against the actual values. This way, one can come up with a probabilistic model to predict the price range. One also needs to inspect how and to what extent the model violates its assumptions

3. How can stock returns be normally distributed?

   It is a standard assumption in financial theory (and models like CAPM, Black-Scholes option pricing, portfolio theory) that returns follow a normal distribution. Empirically also, over many years across different markets and asset classes, there have been numerous studies to show that it is approximately true (generally speaking).

4. How can we convert a daily standard deviation value to annualized standard deviation?

   We multiple the daily standard deviation value with the square root of 252 to get the annualized figure. Here, we assume that there are 252 trading days in a calendar year. Many practitioners multiply the daily value with the square root of 365 as well.

5. How is standard deviation different then volatility?

   Usually, we use standard deviation as a proxy for the historical volatility of an asset.

6. If we have two strategies whose standard deviations are: a) 10% annually, b) 20% for 10-year data. Which strategy has higher volatility?

Before we can compare these values, we need to scale both values similarly. Here, the standard deviation for one strategy has been shown as an annualized number. The standard deviation of the other strategy is shown across 10 years. Here, we should annualize the latter so as to compare them directly.

7. If we have a ten-stock portfolio, then how do we optimize the portfolio?

   We can use a variance-covariance box method to optimize the weights of a portfolio with 'n' securities. We will cover this in Python in one of the upcoming tutorials. Meanwhile, you can read this.

8. Is it optional to use log return instead of percentage returns? Is there a case where I need to use log returns instead of percentage change returns?

   Yes. We can either use log returns or simple returns to evaluate strategies/investments. However, we need to be consistent. Do keep in mind that the cumulative returns are calculated differently for simple and log returns. The reason for this is that simple returns assume periodic compounding, whereas log-returns assume continuous compounding. When using simple returns, the cumulative returns are calculated as the cumulative product of (1+ periodic simple-returns). In the case of log returns, cumulative returns are simply calculated as the cumulative sum of periodic log returns. So calculations with log returns are much easier, and hence log-returns are a preferred choice for many.

9. There is a rare chance that the returns will fall beyond 3SD then why should we consider using 4 and 5 SD?

   Yes, theoretically, 3 SDs values cover up to 99.7% values which means that returns should lie between +/-3 SD around the mean. However, as we know that it is a standard assumption that stock returns follow a normal distribution which does not always hold. This assumption has been violated quite a few times by the market when in turmoil, such as the Great Depression (1929), the coronavirus market crash (2020), etc. These crashes are well beyond 3 SDs, suggesting we should consider modelling up to 4 or 5 SDs.

10. What is the Capital Asset Pricing Model (CAPM)?

    As the name suggests, it is an asset pricing model used to calculate the expected return of a given asset. The example here should give you a good idea of how it is estimated.

